

Short Review: Rate Coding and Motor Unit Recruitment Patterns

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ABSTRACT

It has long been accepted among physiologists that a muscle possesses two mechanisms by which it can increase its tension. One of these mechanisms is termed rate coding and, briefly, it involves the rate at which neural impulses are conducted to the individual motor units comprising the muscle. The second mechanism is motor unit recruitment, referring to the number of available motor units that are actually called into play.

What is not so clear, at this time, is the extent to which these mechanisms interrelate during the generation of muscular tension. This paper reviews the recent literature addressing this topic. The second objective of this paper is to review the literature concerning the actual pattern of motor unit recruitment as the muscle creates tension. It is the purpose of this paper to elucidate the nature by which the neuromuscular system varies and controls the amount of muscular tension generated by a muscle.

KEY WORDS: Rate coding, motor unit, connectivities, recruitment, decreruitment.

Introduction

The human organism is capable of performing a seemingly endless array of physical activities in both everyday life and in athletic endeavors. The same muscles used to execute a very refined movement requiring a minimal amount of force may also be needed to complete a gross motor task that requires a great amount of muscular force. This implies that most muscles comprising the human muscular system feature significant flexibility in the tension they can create. The ability of muscles to vary, or grade, the degree of tension that they generate makes possible the wide variety of motor tasks performed by the human organism.

Two primary mechanisms have been demonstrated to account for a muscle's ability to vary its contractile force. First, an organism may, via the central nervous system (CNS), adjust the firing rate of neural impulses moving down the motoneuron which controls a motor unit (11, 12, 25). This phenomenon is referred to as rate coding (28).

Second, again by way of the CNS, the organism can alter the number of motor units recruited to perform the task at hand. In either situation it is readily apparent that the CNS acts as the command center in determining the amount of force exerted by the muscle, or group of muscles, necessary to perform the particular motor task.

The purposes of this review are twofold: One, to review the relative contributions of rate coding and motor unit recruitment in terms of grading the total amount of tension developed by a muscle; two, to examine the patterns by which motor units are recruited and disengaged as a muscle increases or decreases, respectively, the degree of force with which it contracts. Hopefully, this will provide some insights as to how the muscular system in athletes is able to perform such a wide variety of skills requiring different force production. Greater understanding of the process of motor unit recruitment should enable the coach and/or athlete to design a training program that will optimally stimulate and develop motor units that are recruited and utilized in the performance of a particular athletic event.

Motor Unit Recruitment and Rate Coding

It has been clearly demonstrated that both motor unit recruitment and rate coding play essential roles in the ability to vary the tension produced by a muscle (11, 12, 26, 27). However, until recently there has been little research to ascertain the relative roles that each of these two mechanisms play. Fortunately, there now is some experimental data that shed light on this issue (4, 9, 10, 11, 12).

Before proceeding any further, let us briefly review what is meant when we refer to the processes of motor unit recruitment and rate coding. A single motor unit consists of one motoneuron and all of its associated muscle fibers (21, 25). Any whole muscle may be comprised of as few as 100 motor units or less, or by as many as hundreds of motor units (21, 25). To increase the total tension developed by a muscle, the CNS may recruit additional motor units available within a muscle. Another strategy the CNS may utilize to increase muscular force calls for the motoneurons already activated to increase the frequency of neural impulses con-

ducted. This results in increasing the tension produced by the muscle without recruiting additional motor units (4). This second mechanism has been termed rate coding (28).

The critical question to be resolved is which of these two mechanisms plays a more significant role, and how do they interrelate? The literature indicates that the relative importance of motor unit recruitment and rate coding is determined by the size of the muscle involved, its function and its fiber type composition (24).

Evidence suggests that small muscles such as the adductor pollicis (which is homogenous in its fiber type composition, 72 to 91 percent type I fibers) initially rely upon recruitment to grade their force of contraction (4, 9, 10, 24). In fact, 86 percent of the adductor pollicis' motor units are recruited at 30 percent of its maximal voluntary contraction (MVC), and all of its motor units have been recruited at 50 percent MVC (24). As the adductor pollicis approaches 100 percent of its MVC, additional muscular tension is due to an increase in the rate coding of the motor units already activated. Hence, a sequence is observed where motor unit recruitment initially plays the more prominent role in increased muscular tension, followed by a reliance upon rate coding to produce any further muscular tension (9, 10, 24).

A different phenomenon occurs during the graded contraction of large muscles such as the deltoid and the biceps brachii, both of which are typically comprised of a heterogeneous population of fast twitch (FT) and slow twitch (ST) motor units (9, 10, 24). When these muscles increase their force production, initially there is an increase in the rate coding of low threshold motor units. These low threshold motor units have been determined to be comprised of ST muscle fibers and are innervated by small motor neurons (6, 8). These low threshold motor units are typically the first ones recruited, and then as the muscle continues to increase its level of tension we see the recruitment of additional motor units of increasingly higher thresholds (4, 9, 10, 24). In the deltoid, this process continues up to 88 percent of MVC. At this point, any further increase in force production must be accounted for solely by an increase in rate coding.

The interplay between motor unit recruitment and rate coding in grading the contraction of any particular muscle can best be explained by the work of Bigland and Lippold (4). Each motor unit has a critical firing level at which it can initiate a twitch or contraction. Low threshold motor units are recruited with a slower frequency of neural impulses than higher threshold motor units. But, for any motor unit there is a limitation at which greater impulse frequency causes an increase in tension developed by that motor unit (4). At this point, tetanic fusion occurs and any further increase in impulse frequency will be wasteful and useless. It has been shown that in low threshold motor units of the cat's gastrocnemius, this fusion can occur at a stimulus frequency of 10 neural impulses per second, and that these motor units develop maximal tension at 20 impulses per second (37). Thus, any increase in the frequency of impulses can cause greater muscle tension only by the recruitment of higher threshold motor units.

Interestingly, it has been shown that low threshold motor units operate within a greater range of impulse frequency than high threshold motor units (4, 9, 10). This suggests that when a heterogeneous muscle gradually increases the amount of tension, low threshold motor units are responsible for a greater contribution to MVC than high threshold motor units. In fact, in experiments done with the gastrocnemius of the cat, at a stimulus rate of 20 impulses per second, the muscle reaches 70 percent MVC (37). This data does not downplay the importance of high threshold motor units in the contraction of muscle. Indeed, a muscle cannot reach MVC without recruiting its high threshold motor units. The importance in the recruitment of these high threshold, FT motor units lies in the fact that although they sometimes do not develop maximum tension until the stimulus frequency reaches 100 per second, they are capable of generating more force than low threshold ST motor units (12, 25). But, when examining the effect that neural impulse frequency plays on motor unit recruitment pattern in a heterogeneous muscle, it is evident that the impulse frequencies at which motor units are capable of generating tension form an S-shaped curve. That is, motor units are capable of developing more tension when the frequency of the stimulus is high, until tetanus is achieved. But when the tension created by a large muscle such as the biceps brachii is very great or very little, there are very few changes in the number of motor units called into play. These results demonstrate that in large heterogeneous muscles such as the deltoid or the biceps brachii, the gradation of a muscle's force is due to a combination of motor unit recruitment and rate coding, except when it is operating at near 0 percent MVC or 100 percent MVC. At these extremes, rate coding plays a more critical role (3, 4).

At this time, scientists do not have enough evidence to state with any real confidence whether it is the muscle fiber composition, function of the muscle, or the size of the muscle that dictates the relative importance of rate coding and motor unit recruitment to the gradation of the muscle's tension. It does seem reasonable that muscle fiber composition plays a key role. This would explain why a muscle containing a homogenous population of low threshold ST motor units would recruit all of its motor units very easily and then rely on rate coding to cause any further increase in its tension. However, with a muscle comprised of a heterogeneous population of ST and FT motor units, there is an increase in motor unit recruitment of up to 88 percent MVC (24). This indicates that the fiber composition of a muscle and the critical firing levels associated with different fiber types, determine the interrelationship of motor unit recruitment and rate coding in the gradation of muscular force.

According to Desmedt and Godaux (11, 12), the recruitment patterns of motor units in a contraction muscle remain the same regardless of the nature of the contraction; i.e. a slow ramp contraction or a fast ballistic contraction. But, during ballistic contractions there does appear to be some neural potentiation of the higher threshold motor units without a commensurate potentiation of the lower threshold motor units. The potentiation of the high thresh-

old motor units results in a decreased critical firing level of these motor units and an earlier recruitment of them (11). The potentiation of high threshold motor units that occurs during the ballistic contractions, but not ramp contractions, suggests that muscles comprised of a heterogeneous population of fiber types might be capable of generating more force during ballistic contractions than would be possible during slow graded contractions.

There still exists a need for more research on the interrelationship between motor unit recruitment and rate coding in the gradation of muscular tension. However, it is safe to say at this juncture that the relative roles that these two processes play in varying the force developed by a muscle is related to the muscle's fiber type composition, its function and its size (24).

Motor Unit Recruitment Pattern

The focus of this paper will now be turned to the manner in which motor units are recruited during gradation of the muscle's force output. A large body of evidence indicates that there is a preferential sequence by which the motor units comprising a muscle are called into play. This sequence has been described by the "size principle" of motor unit recruitment (18, 21).

The leading proponents of the size principle are Elwood Henneman and H. Peter Clamann. These two scientists have conducted numerous studies which seem to substantiate the existence of the size principle of motor unit recruitment. In brief, the size principle states that the size of the motoneuron is related to the size of the motor unit and diameter of the axon of that particular motoneuron. In turn, the axon's diameter determines the velocity at which the motoneuron is able to conduct neural impulses: this velocity is related to the type of muscle fibers innervated by the motoneuron (8, 18, 36). The size of the motoneuron is directly related to the size of its soma. This is important because the size of the soma is responsible for the critical firing level of the motoneuron. This means that a small motoneuron, featuring a small soma, will have a low critical firing level and can thus be categorized as a low threshold motor unit. Conversely, a large motoneuron, possessing a large soma, innervates a high threshold motor unit.

Let us now take an opportunity to further elaborate upon the means by which the size principle operates. As an excitatory post synaptic potential, or impulse, arrives at the soma of the motoneuron, it must spread passively to the trigger zone at the neuron's axon hillock region. Along its trek the impulse encounters resistance which tends to decrease the strength, or amplitude, of the impulse. If the amplitude of the impulse has been reduced too much by the time it arrives at the trigger zone, it will not be strong enough to reach the threshold of the neuron. If this is the case, no action potential will be initiated in that neuron and no impulse will be conducted down the axon to the neuromuscular junction (21). It is also known that the size of the neuron's soma is directly related to the amount of resistance

encountered by the postsynaptic potential as it makes its way toward the trigger zone. Hence, larger motoneurons with their larger somas have higher thresholds, as it is difficult for the postsynaptic potential to maintain an amplitude adequate to induce an action potential to be fired at the trigger zone. A small motoneuron, on the other hand, has a small soma which decreases the amount of resistance the postsynaptic potential must overcome on its way to the trigger zone. This increases the likelihood that the stimulus will be of sufficient strength to reach the threshold level and initiate an action potential in the neuron which will then be propagated toward the neuromuscular junction (21). As an example of these principles, it has been shown that high threshold motor units in the soleus of the cat could only be recruited when the frequency of the impulses were 40 per second. At this rate the impulses, which individually are below threshold level, can sum, or add to each other, and reach threshold level causing an action potential to be fired. Alternatively, low threshold motor units could be recruited when the frequency of impulses was as low as five per second (26, 37). These differences in frequency rates necessary to reach threshold of the motor units reflect the resistance posed by the soma of the particular motor unit's motoneuron. According to the size principle then, it can be said that as the CNS recruits more motor units to increase the tension developed by the muscle, it begins with the smallest motor units and progressively works up to the largest motor units. The disengagement of motor units, necessary to decrease the tension developed by a muscle, also adheres to the size principle. That is the largest motor units, which are the last to be recruited as the force of the muscle increases, are the first to stop contracting as the force developed by the muscle decreases. It can be said that the disengagement sequence is the exact inverse of the recruitment sequence of motor units (9, 10).

A vital premise of the size principle is that all of the motoneurons of the motoneuron pool of a muscle receive the same stimulus from the CNS (21). If true, this factor lends further credibility to the existence of the size principle of motor unit recruitment. When looking at all of the motor units found in a muscle, the smallest ones will be recruited first and the largest motor units will be recruited last. Only a very high frequency of excitatory postsynaptic potentials will result in the recruitment of large motor units.

Henneman et al (7, 8, 18) defined the size principle quite eloquently. He believes the size principle is the only way in which the organism can ensure that the degree of force developed by a muscle is appropriate for the task at hand. If not for the efficacy of the size principle, muscle systems would constantly evoke inappropriate movements as a result of creating too much force or too little force necessary to execute a specific motor task. If the size principle did not hold true, every movement would involve an inefficient process of guesswork to estimate how much force was needed to perform a task. The size principle dictates that the smallest, least powerful motor units are recruited first and that progressively larger, more powerful motor units

are recruited only when necessary.

Henneman also states that the size principle ensures the most metabolically efficient means of completing any given movement. This is due to the fact that high threshold motor units, relative to low threshold motor units, are very costly metabolically due to their low oxidative capacities. The size principle guarantees that these large, high threshold motor units will only be recruited after the smaller, more metabolically efficient motor units have been recruited. In response to those who suggest that afferent impulses determine the sequence of motor unit recruitment, Henneman believes that to rely upon afferent feedback would be ineffective in that it would require too much time for the CNS to process this afferent information and then decide upon an appropriate recruitment pattern for the motor task at hand (18).

In addition to Henneman and his colleagues, there are other researchers who have accumulated substantial evidence to support the size principle. For example, it has been demonstrated that the size principle applies to both reflex activity and voluntary contractions of a muscle (6). It has also been shown that the size principle applies to ballistic contractions as well as slow ramp contractions of the same muscle (11, 12). Additionally, it has been illustrated that the fiber composition of a motor unit does not seem to alter the recruitment pattern of a muscle's motor units. Indeed, both large muscles featuring a heterogeneous distribution of FT and ST motor units and small muscles comprised of a homogenous distribution of ST motor units seem to conform to the size principle of recruitment (2, 9, 10).

Given the abundance of evidence supporting the existence of the size principle of motor unit recruitment, few scientists find fault with it when looking at an isolated muscle performing a movement in one direction or single plane. However, most motor tasks are not carried out under such conditions. For example, most movements involve several muscles contracting simultaneously. Also, the same muscle can contract in more than one direction in order to perform different motor tasks. A muscle that functions as a prime mover in one movement may serve as a synergist in another movement. Bear in mind that motoneurons receive not only excitatory, but also inhibitory impulses from the descending pathways of the CNS. During complex motor tasks, the neurons also receive afferent feedback which is relayed to the higher levels of the CNS, and can have some impact on the recruitment pattern employed during the performance of the movement.

As appealing as Henneman's size principle may seem in explaining how motor units are recruited, it does seem too simplistic given the different functions a muscle may perform and the different directions it can contract in. In fact, there are several prominent scientists who take issue with the size principle in its attempt to explain the recruitment pattern of any particular muscle regardless of the conditions it is working under. These scientists have questioned the viability of the size principle for different reasons. The following is a review of some of these arguments.

Garnett and Stephens (16) have completed experiments

which indicate that the recruitment pattern of a muscle can be at least partly determined by afferent input from the cutaneous tissue surrounding a muscle. They have found that by electrically stimulating the cutaneous tissue, the critical firing levels of motor units can be altered. Specifically, the low threshold, small motor units are inhibited while the high threshold, large motor units are excited. The result is a change in the recruitment pattern where we see high threshold motor units being recruited before low threshold motor units.

The work of Desmedt and Godaux (13) also contradicts the size principle in certain situations. They have demonstrated that the recruitment pattern of the first dorsal interosseous muscle changes rather dramatically depending upon the direction in which it is contracting and whether it is acting as a prime mover or a synergist. After completing a statistical analysis, it was observed that some motoneurons changed their critical firing levels by as much as two standard deviations when the muscle switched its role from being the prime mover to being a synergist. According to Desmedt and Godaux, the reason for the changes in these critical firing levels is the difference in connectivities within the motoneuron pool. The term connectivities refers to the manner in which any one neuron makes synaptic contacts upon other neurons and is influenced by synaptic contacts from other neurons. These connectivities are dependent upon the direction that the muscle contracts in. As a result, there are different recruitment patterns related to the different directions in which the muscle is able to contract.

The speed at which a muscle contracts also may affect the recruitment pattern of that muscle (35). There is evidence to suggest that at extremely high rates of muscular contraction there exists some form of neural switching that allows the high threshold, large motor units to be preferentially recruited before low threshold, small motor units. It has been postulated that the reason high threshold motor units are recruited first in ballistic contractions is that the synergists which are co-contracting increase the afferent feedback received by the CNS, and this in turn affects the recruitment pattern of the prime mover (16, 17). Others, who have demonstrated the plasticity of recruitment patterns for most muscles, believe that the input received by the motoneuron pool from the CNS is task specific. If task specificity does exist, then different motor tasks would receive different information from the descending levels of the CNS, dictating the optimal recruitment pattern for the task at hand.

Of all the arguments brought forth to dispute the viability of the size principle of recruitment, the most intriguing is the one advanced by Person (31). Henneman's size principle is predicated on his belief that all of the motoneurons within a motoneuron pool receive exactly the same amount and type of stimulation from the CNS. However, Person believes that this is not the case. Rather, he believes that experimental evidence indicates that each individual motoneuron within the pool has its own preferred sources of excitation and inhibition. If this is true, recruitment patterns are task specific and are controlled by different cortical levels depending upon the nature of the task and its famili-

arity to the organism. Person believes that when a muscle contracts in a fixed, isolated position, the size principle is applicable. But when the muscle contracts in a free posture the recruitment pattern varies. Person proposes that during most motor tasks each motoneuron within the motoneuron pool acts individually and that recruitment patterns are affected by afferent feedback and the coordination of the higher levels of the CNS. Thus, the number of recruitment patterns a muscle can exhibit is limited only by the number of motor tasks it can participate in.

In summary, there is convincing evidence to support the existence of the size principle of motor unit recruitment: however, this concept does have restrictions and must only be utilized within its proper context. The size principle can be verified in controlled, isolated instances when a muscle performs a simple, unidirectional motor movement. The size principle may also be appropriate for muscles that are comprised of a homogenous distribution of muscle fiber types.

However, in more natural settings where muscles can contract in different directions, where synergists may co-contract with the prime mover, where there is a significant amount of afferent feedback, where the function of the muscle changes from that of the prime mover to that of a synergist, the recruitment pattern of motor units cannot be unquestioningly ascribed to the size principle. Furthermore, it must be pointed out that some motor tasks, by their very nature, belie the constancy of the size principle. Some explosive, multidirectional movements employing many muscle groups necessitate the immediate recruitment of the more powerful, high threshold, FT motor units regardless of the metabolic cost in doing so.

This is not to suggest that the size principle is no longer a viable concept and should be discarded. Rather, it must be examined and utilized within the appropriate limitations. Upon consideration of the great variety of movements that most muscles can take part in, it simply does not seem reasonable that a concept such as the size principle would hold true in all circumstances. In fact, to blindly embrace the size principle would be to grossly oversimplify the neuromuscular system, a system which inherently defies simplification.

Understanding the process by which motor units are recruited may directly affect the limitations of any weight training program. In light of the research supporting the size principle of motor unit recruitment, it is not possible to design resistance exercises that develop large, fast twitch motor units exclusively. Alternatively, the fact that the size principle does not apply to all types of movement sets limitations to how closely resistance exercises can mimic movements that commonly occur in some dynamic sport activities. If, in fact, recruitment patterns are altered by factors such as afferent feedback and changes in direction, it would be difficult to design a weight training program that accurately mimics all the movements seen in a particular sports activity. These facts prove that while weight training programs can improve sports performance, it is simplistic to believe that weight training by itself will develop superior athletic ability. A properly designed resistance training pro-

gram serves as a single component of an overall training program for the athlete.

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